

Evaluating Cross-Validation Reliability in Kaggle Tabular Binary Classification Competitions Using Bayesian Melding

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Abstract—Cross-validation is one of the most widely used methods for estimating machine learning model performance before deployment or external evaluation. In competitive machine learning platforms such as Kaggle, practitioners commonly rely on cross-validation scores and public leaderboard feedback to guide model development, while final model performance is determined using a hidden private leaderboard. This creates an important challenge in determining whether cross-validation estimates reliably predict true generalisation performance.

This study investigates the reliability of cross-validation in Kaggle tabular binary classification competitions using an empirical Bayesian melding framework. Three completed Kaggle competitions were analysed: Santander Customer Transaction Prediction, Porto Seguro’s Safe Driver Prediction, and Homesite Quote Conversion. For each competition, 1,000 controlled model configurations were trained and evaluated using stratified 5-fold cross-validation, resulting in 3,000 cross-validation observations. Due to Kaggle leaderboard submission constraints, a representative subset of 100 models per competition was selected across model families and performance ranges, producing 300 models with cross-validation, public leaderboard, and private leaderboard scores.

The results show very strong alignment between cross-validation AUC and private leaderboard AUC across all competitions, with within-competition centred Pearson correlation of 0.9958. Public leaderboard scores were also highly correlated with private leaderboard scores, with centred Pearson correlation of 0.9988. Bayesian melding was used to treat cross-validation, public leaderboard, and private leaderboard scores as noisy observations of latent true model performance. The estimated global cross-validation bias was small at 0.00042 AUC, while public leaderboard bias was 0.00057 AUC. These findings suggest that, when carefully designed and evaluated across model families and competitions, stratified cross-validation can provide a highly reliable estimate of hidden leaderboard performance in tabular binary classification competitions.

Index Terms—cross-validation, Bayesian melding, Kaggle, leaderboard, model validation, binary classification, tabular data, generalisation, machine learning evaluation

I. INTRODUCTION

Machine learning models are only useful when their measured performance reflects how well they generalise to unseen data. In practice, this is difficult to guarantee. During model development, practitioners often estimate model quality using validation methods such as holdout validation

or cross-validation. However, final model performance may differ from validation estimates due to sampling noise, data leakage, model selection bias, distribution shift, or overfitting to validation feedback.

This issue is especially important in competitive machine learning environments such as Kaggle [1]. Kaggle competitions provide participants with training data and a hidden test set. Participants submit predictions for the test set and receive feedback through a public leaderboard. The final ranking is usually determined by a private leaderboard, which is hidden until the competition ends. As a result, practitioners typically make modelling decisions using two imperfect sources of information: local cross-validation scores and public leaderboard feedback [2], [3]. Neither source is guaranteed to perfectly represent private leaderboard performance.

Cross-validation is widely used because it allows the training data to be split into multiple folds, giving a more stable estimate of model performance than a single validation split. However, cross-validation can still be biased [4], [5]. It may overestimate performance when preprocessing or feature selection leaks information across folds, when hyperparameters are repeatedly tuned against the same validation strategy, or when validation folds do not represent the hidden test distribution. Public leaderboard feedback can introduce another form of bias because users may repeatedly submit models and adjust their approach based on leaderboard movements.

This research studies the reliability of cross-validation in Kaggle tabular binary classification competitions. The focus is not on achieving the best Kaggle rank, but on understanding whether cross-validation scores are trustworthy estimates of hidden private leaderboard performance. This is valuable because practitioners often use competition results, benchmark scores, and validation pipelines to make decisions about model quality.

The study uses three completed Kaggle tabular binary classification competitions: Santander Customer Transaction Prediction, Porto Seguro’s Safe Driver Prediction, and Homesite Quote Conversion [6]–[8]. For each competition, 1,000 controlled model configurations were trained using six model families: LightGBM, XGBoost, CatBoost, Random Forest,

ExtraTrees, and Logistic Regression. Each model was evaluated using stratified 5-fold cross-validation. A representative subset of 100 models per competition was then evaluated using Kaggle public and private leaderboard scores.

To integrate these evaluation signals, this research adopts a Bayesian melding framework [9]. Each model is assumed to have an unobserved true generalisation performance. Cross-validation, public leaderboard, and private leaderboard scores are treated as noisy observations of that latent performance. This approach allows validation bias, uncertainty, and relative evaluation-source reliability to be quantified.

This study makes three main contributions. First, it constructs a controlled empirical evaluation dataset containing 3,000 trained model configurations across three completed Kaggle tabular binary classification competitions. Second, it collects a representative leaderboard-scored subset of 300 models with cross-validation, public leaderboard, and private leaderboard scores, enabling direct analysis of validation reliability against hidden leaderboard outcomes. Third, it applies Bayesian melding to model-validation analysis by treating cross-validation, public leaderboard, and private leaderboard scores as noisy observations of latent true model performance. This provides both correlation-based evidence and uncertainty-aware estimates of validation bias and reliability.

II. RELATED WORK

Machine learning model evaluation has been widely studied in relation to generalisation, validation bias, benchmark reliability, and leaderboard feedback. These issues are central to competitive machine learning, where participants rely on validation scores and leaderboard signals to decide which models to submit and trust.

A major concern in machine learning competitions is leaderboard overfitting. Roelofs et al. conducted a large-scale empirical study of Kaggle competitions and investigated whether repeated submissions lead to overfitting to the public leaderboard [2]. Their work found limited evidence of severe adaptive overfitting in many competitions, although discrepancies between public and private leaderboard scores can still occur when test sets are small, noisy, or not representative.

Leaderboard design has also been studied as a way to reduce overfitting. Blum and Hardt proposed the Ladder mechanism, which only reveals leaderboard improvements when a submission achieves a statistically meaningful gain over previous submissions [3]. Chaibub Neto et al. later proposed a bootstrapped leaderboard mechanism to reduce overfitting in challenge-based competitions [10].

The broader problem of adaptive data analysis provides a theoretical foundation for understanding leaderboard overfitting. Dwork et al. showed that repeatedly querying the same validation data can invalidate standard statistical assumptions and lead to overfitting [11]. Their reusable holdout framework demonstrates that validation data can be reused more safely if information leakage is controlled [12].

Cross-validation bias is another important issue. Varma and Simon showed that using cross-validation both for model se-

lection and performance estimation can produce optimistically biased results [4]. Cawley and Talbot similarly argued that model selection itself can overfit when validation scores are noisy [5]. Krstajic et al. also highlighted practical pitfalls that arise when cross-validation is used for model selection and assessment without careful experimental design [13]. Data leakage is another common cause of inflated validation performance [14], [15]. Although this study uses consistent preprocessing pipelines to reduce leakage risk, these works show why validation estimates must be interpreted carefully.

Benchmark generalisation research also questions whether fixed test sets fully represent real-world performance. Recht et al. showed that models evaluated on newly collected ImageNet-style data experienced performance drops compared with the original ImageNet benchmark, even though model rankings were relatively stable [16]. A similar question applies to Kaggle competitions: cross-validation may preserve model ranking but still differ from private leaderboard performance.

Although prior work has studied leaderboard overfitting and validation bias, fewer studies directly compare cross-validation, public leaderboard, and private leaderboard scores across a controlled population of many model configurations and multiple tabular binary classification competitions. This study addresses that gap by combining large-scale empirical experimentation with Bayesian melding.

III. RESEARCH QUESTIONS

The primary research question is:

RQ1: How reliable is stratified 5-fold cross-validation AUC as an estimate of private leaderboard AUC in Kaggle tabular binary classification competitions?

The secondary research questions are:

- **RQ2:** How closely do public leaderboard AUC scores align with private leaderboard AUC scores?
- **RQ3:** Does validation reliability differ across competitions and model families?
- **RQ4:** What systematic bias and noise are associated with cross-validation and public leaderboard estimates?
- **RQ5:** Can Bayesian melding provide a useful framework for integrating cross-validation, public leaderboard, and private leaderboard scores into a unified estimate of latent model performance?

IV. METHODOLOGY

This study uses an empirical experimental design to evaluate the reliability of cross-validation in Kaggle tabular binary classification competitions. The methodology consists of five stages: competition selection, large-scale model generation, cross-validation evaluation, representative leaderboard scoring, and Bayesian melding analysis.

A. Competition Selection

Three completed Kaggle competitions were selected: Santander Customer Transaction Prediction, Porto Seguro's Safe Driver Prediction, and Homesite Quote Conversion [6]–[8]. All three competitions are tabular binary classification tasks with public and private leaderboard evaluation.

Experimental Workflow

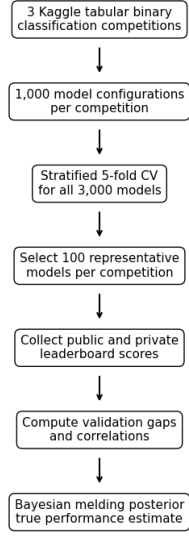


Fig. 1. Experimental workflow showing the construction of the 3,000-model cross-validation population, selection of 300 representative leaderboard-scored models, and Bayesian melding analysis.

B. Experimental Dataset Design

The final experiment contains two dataset layers. The first layer is the full cross-validation population, consisting of 1,000 trained model configurations per competition and 3,000 cross-validation observations in total. The second layer is the leaderboard-scored subset. Due to Kaggle submission limits and anti-abuse throttling, public and private leaderboard scores were collected for a representative subset of 100 models per competition, producing 300 complete leaderboard-scored observations.

C. Model Families and Configuration Generation

Six model families were used: LightGBM [17], XGBoost [18], CatBoost [19], Random Forest [20], ExtraTrees [21], and Logistic Regression. For each competition, 1,000 model configurations were generated using controlled random sampling of hyperparameters. The configuration distribution was kept consistent across competitions: 350 LightGBM models, 350 XGBoost models, 150 CatBoost models, 75 Random Forest/ExtraTrees models, and 75 Logistic Regression models.

D. Preprocessing and Cross-Validation

All models were evaluated using stratified 5-fold cross-validation. Stratification was used because all three tasks are binary classification problems, and class imbalance can affect validation reliability if folds are not class-balanced.

For tree-based and boosting models, missing values were handled using median imputation. Logistic Regression used median imputation followed by standard scaling. The primary cross-validation metric was area under the receiver operating

characteristic curve (ROC-AUC). For Santander and Home-site, Kaggle leaderboard scores were reported as AUC. Porto Seguro uses Normalized Gini as its official leaderboard metric, which was converted to AUC using:

$$\text{AUC} = \frac{\text{Gini} + 1}{2}. \quad (1)$$

E. Representative Leaderboard Subset Selection

For each competition, 100 representative models were selected from the 1,000 trained configurations for leaderboard evaluation. The subset was selected across model families and across low, median, and high cross-validation performance ranges. This avoided selecting only the strongest models and reduced cherry-picking risk.

The final per-competition leaderboard subset consisted of 25 LightGBM models, 25 XGBoost models, 20 CatBoost models, 15 Logistic Regression models, and 15 Random Forest/ExtraTrees models combined.

F. Validation Reliability Metrics

Validation reliability was assessed using two gap metrics:

$$\Delta_{\text{CV},m} = \text{CV}_m - \text{PrLB}_m, \quad (2)$$

$$\Delta_{\text{PLB},m} = \text{PLB}_m - \text{PrLB}_m, \quad (3)$$

where CV_m is cross-validation AUC, PLB_m is public leaderboard AUC, and PrLB_m is private leaderboard AUC for model m .

The study also computed Pearson and Spearman correlations between cross-validation AUC and private leaderboard AUC, and between public leaderboard AUC and private leaderboard AUC. Correlations were computed overall, by competition, by model family, and using a within-competition centred approach.

G. Bayesian Melding Framework

Bayesian melding was used to integrate cross-validation, public leaderboard, and private leaderboard scores as noisy observations of an underlying latent model performance parameter. For each model m , the latent true generalisation performance is denoted by θ_m . The observed evaluation signals are:

$$\mathbf{x}_m = \{\text{CV}_m, \text{PLB}_m, \text{PrLB}_m\}. \quad (4)$$

The Bayesian melding objective is to estimate:

$$p(\theta_m \mid \text{CV}_m, \text{PLB}_m, \text{PrLB}_m). \quad (5)$$

A weak Gaussian prior is placed over latent model performance:

$$\theta_m \sim \mathcal{N}(\mu_0, \tau_0^2). \quad (6)$$

The observation model is:

$$\text{CV}_m = \theta_m + b_{\text{CV}} + \epsilon_{\text{CV}}, \quad (7)$$

$$\text{PLB}_m = \theta_m + b_{\text{PLB}} + \epsilon_{\text{PLB}}, \quad (8)$$

$$\text{PrLB}_m = \theta_m + \epsilon_{\text{Pr}}. \quad (9)$$

TABLE I
FULL CROSS-VALIDATION POPULATION BY MODEL FAMILY

Model family	Number of models
LightGBM	1050
XGBoost	1050
CatBoost	450
Logistic Regression	225
ExtraTrees	126
Random Forest	99
Total	3000

Equivalently, the likelihood terms are:

$$p(\text{CV}_m | \theta_m) = \mathcal{N}(\theta_m + b_{\text{CV}}, \sigma_{\text{CV}}^2), \quad (10)$$

$$p(\text{PLB}_m | \theta_m) = \mathcal{N}(\theta_m + b_{\text{PLB}}, \sigma_{\text{PLB}}^2), \quad (11)$$

$$p(\text{PrLB}_m | \theta_m) = \mathcal{N}(\theta_m, \sigma_{\text{Pr}}^2). \quad (12)$$

Using the observed evaluation vector $\mathbf{x}_m$, the posterior is proportional to the prior multiplied by the likelihood terms:

$$\begin{aligned} p(\theta_m | \mathbf{x}_m) &\propto p(\theta_m) p(\text{CV}_m | \theta_m) \\ &\quad \times p(\text{PLB}_m | \theta_m) \\ &\quad \times p(\text{PrLB}_m | \theta_m). \end{aligned} \quad (13)$$

Bias terms were estimated empirically as:

$$b_{\text{CV}} = \text{mean}(\text{CV AUC} - \text{Private AUC}), \quad (14)$$

$$b_{\text{PLB}} = \text{mean}(\text{Public AUC} - \text{Private AUC}). \quad (15)$$

Noise terms were estimated as:

$$\sigma_{\text{CV}} = \text{std}(\text{CV AUC} - \text{Private AUC}), \quad (16)$$

$$\sigma_{\text{PLB}} = \text{std}(\text{Public AUC} - \text{Private AUC}). \quad (17)$$

The private leaderboard uncertainty was assigned as half of the smaller estimated noise term. After bias correction, the posterior true performance estimate was computed using inverse-variance weighting:

$$\theta_m^* = \frac{w_{\text{CV}} \text{CV}_m^{\text{corr}} + w_{\text{PLB}} \text{PLB}_m^{\text{corr}} + w_{\text{Pr}} \text{PrLB}_m}{w_{\text{CV}} + w_{\text{PLB}} + w_{\text{Pr}}}, \quad (18)$$

where each weight is the inverse of the corresponding variance:

$$w = \frac{1}{\sigma^2}. \quad (19)$$

V. RESULTS

A. Full Cross-Validation Population

The full experimental population consisted of 3,000 trained model configurations, with 1,000 models evaluated for each competition. Table I summarises the model-family distribution.

The cross-validation distributions varied strongly across competitions. Homesite had the highest overall AUC range, Santander achieved private-leaderboard-level performance around the mid-to-high 0.80s, and Porto Seguro was lower in absolute AUC after converting Normalized Gini to AUC. Fig. 2 shows the full CV distribution by competition.

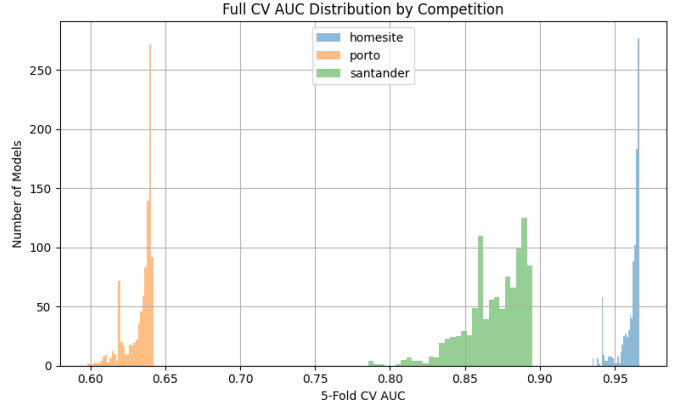


Fig. 2. Distribution of 5-fold cross-validation AUC across the full 3,000-model population.

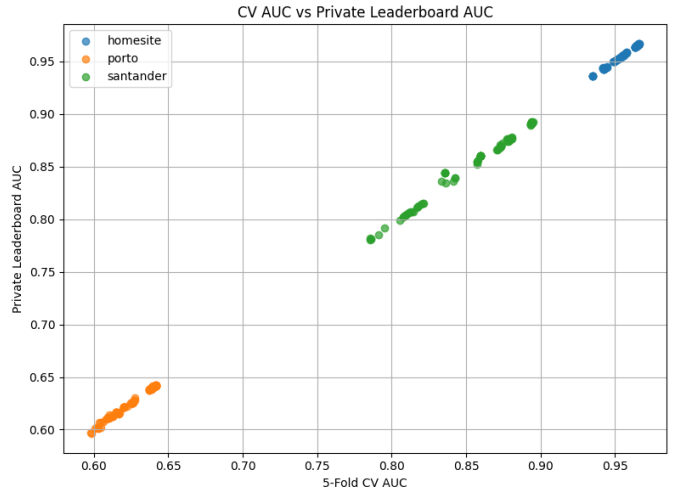


Fig. 3. Relationship between cross-validation AUC and private leaderboard AUC across the 300 leaderboard-scored models.

B. Leaderboard-Scored Subset

The leaderboard-scored dataset contained 300 models, with 100 models per competition. Table II summarises the leaderboard-scored subset by competition.

Homesite showed the smallest validation discrepancy, with a mean absolute CV-private gap of 0.000573 AUC. Porto also showed a small gap of 0.000999 AUC. Santander had the largest discrepancy, with a mean absolute CV-private gap of 0.003576 AUC. Even in Santander, however, the average discrepancy remained below 0.004 AUC, suggesting that stratified cross-validation remained informative despite competition-specific variation.

C. Correlation Between Validation and Private Leaderboard Performance

Table III reports Pearson and Spearman correlations between cross-validation AUC and private leaderboard AUC, as well as between public leaderboard AUC and private leaderboard AUC.

TABLE II
LEADERBOARD-SCORED SUMMARY BY COMPETITION

Competition	Models	Mean CV AUC	Mean Public AUC	Mean Private AUC	Mean Abs. CV-Private Gap	Mean Abs. Public-Private Gap
Homesite	100	0.955840	0.956981	0.956401	0.000573	0.000585
Porto	100	0.626901	0.625845	0.627521	0.000999	0.001924
Santander	100	0.858434	0.858785	0.855992	0.003576	0.002799

TABLE III
CORRELATION BETWEEN EVALUATION SIGNALS

Group	Models	CV-Private Pearson	CV-Private Spearman	Public-Private Pearson	Public-Private Spearman
Overall	300	0.999849	0.998848	0.999919	0.999245
Homesite	100	0.999342	0.987109	0.999481	0.995439
Porto	100	0.996967	0.989533	0.997015	0.992010
Santander	100	0.995336	0.992252	0.999158	0.992158
Within-competition centred	300	0.995804	0.994294	0.998821	0.996460

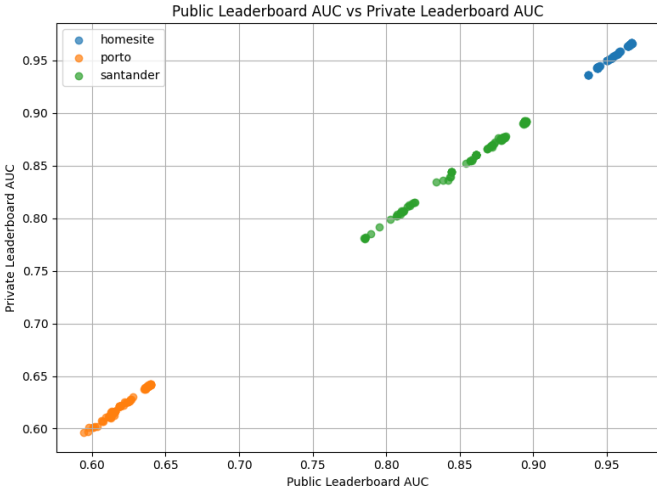


Fig. 4. Relationship between public leaderboard AUC and private leaderboard AUC across the 300 leaderboard-scored models.

The overall CV-private Pearson correlation was 0.999849, while the public-private Pearson correlation was 0.999919. Because overall correlations can be inflated by different competition score ranges, a within-competition centred analysis was also conducted. After centring scores within each competition, CV-private Pearson correlation remained high at 0.995804, and public-private Pearson correlation remained 0.998821. This indicates that the observed reliability is not merely an artefact of combining competitions with different baseline AUC ranges, but also holds within individual competition score distributions.

D. Validation Gap Analysis

Validation gap analysis measures whether cross-validation and public leaderboard scores systematically overestimate or underestimate private leaderboard performance. Table IV summarises the gap metrics by competition.

Homesite and Porto showed slightly negative mean CV-private gaps, indicating that cross-validation was mildly pes-

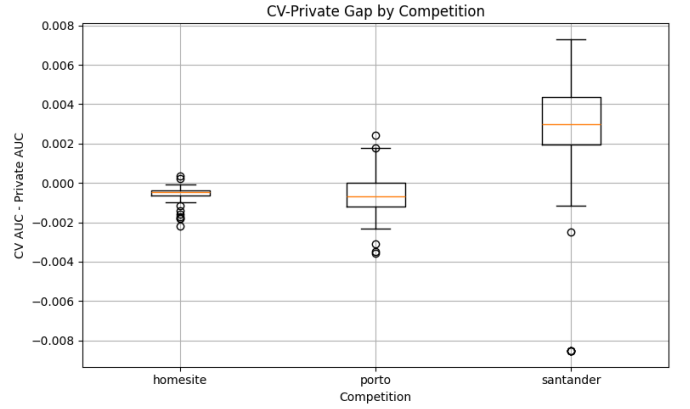


Fig. 5. Distribution of CV-private leaderboard gaps by competition.

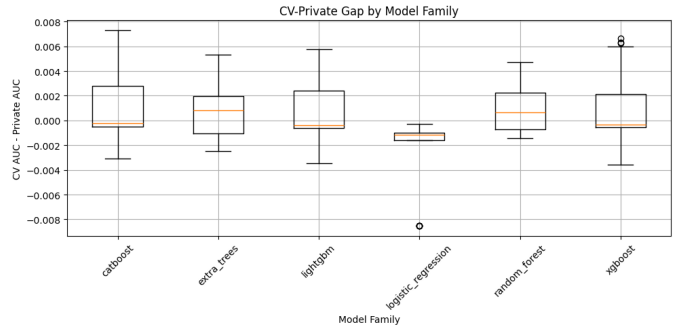


Fig. 6. Distribution of CV-private leaderboard gaps by model family.

simistic on average. Santander showed a positive mean CV-private gap of 0.002442 AUC, indicating mild optimism.

E. Bayesian Melding Results

Bayesian melding was applied to integrate cross-validation, public leaderboard, and private leaderboard scores as noisy observations of latent true model performance. The estimated global CV bias was 0.000420 AUC, while the estimated public leaderboard bias was 0.000566 AUC. The estimated noise term

TABLE IV
GAP SUMMARY BY COMPETITION

Competition	Mean CV-Private Gap	Std CV-Private Gap	Mean Abs. CV-Private Gap	Mean Public-Private Gap	Mean Abs. Public-Private Gap
Homesite	-0.000562	0.000392	0.000573	0.000580	0.000585
Porto	-0.000619	0.001075	0.000999	-0.001676	0.001924
Santander	0.002442	0.003249	0.003576	0.002793	0.002799

TABLE V
BAYESIAN MELDING BIAS AND NOISE ESTIMATES

Parameter	Estimate
CV bias	0.000420
Public leaderboard bias	0.000566
σ_{CV}	0.002446
σ_{Public}	0.002138
$\sigma_{Private}$	0.001069

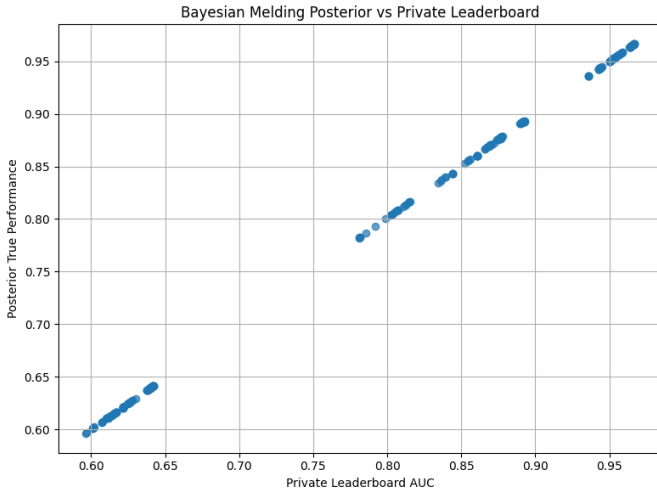


Fig. 7. Bayesian melding posterior true performance estimates compared with private leaderboard AUC.

for cross-validation was $\sigma_{CV} = 0.002446$, while the estimated noise term for the public leaderboard was $\sigma_{Public} = 0.002138$.

The posterior mean remained close to the private leaderboard mean for each competition, indicating that the Bayesian melding procedure produced fused estimates consistent with the strongest available proxy for generalisation. This supports the use of Bayesian melding not as a replacement for leaderboard evaluation, but as an interpretable framework for combining multiple validation signals while explicitly estimating bias and uncertainty.

VI. DISCUSSION

The results indicate that stratified 5-fold cross-validation was highly reliable in the three Kaggle tabular binary classification competitions studied. Across the 300 leaderboard-scored models, cross-validation AUC was strongly correlated with private leaderboard AUC. This relationship remained strong even after scores were centred within each competition,

showing that the finding was not simply caused by combining competitions with different baseline AUC ranges.

The small mean CV-private gap suggests that cross-validation did not systematically misestimate private leaderboard performance in a large way. The global estimated CV bias was only 0.000420 AUC, which is practically small. However, the direction and magnitude of the gap differed by competition. Homesite and Porto showed slightly pessimistic cross-validation estimates, while Santander showed mild optimism. This suggests that validation reliability depends on dataset characteristics, hidden test split, model family, and competition design.

Public leaderboard scores were slightly more aligned with private leaderboard scores than cross-validation scores. This is expected because both public and private leaderboard scores are based on Kaggle hidden test data. However, the difference between public-private and CV-private reliability was small. This finding suggests that a carefully designed local cross-validation strategy can approximate hidden leaderboard behaviour closely, even without relying heavily on public leaderboard feedback.

A practical implication is that cross-validation should not be judged only by whether it exactly matches a hidden leaderboard score. Instead, its usefulness depends on whether it preserves model ordering, produces small absolute validation gaps, and remains stable across model families. In this study, stratified 5-fold cross-validation satisfied these conditions across the analysed competitions. However, the larger Santander gap shows that validation reliability remains dataset-dependent, and practitioners should still inspect validation gaps, leaderboard stability, and possible distribution differences before relying on a single validation score.

A. Limitations

This study has several limitations. First, leaderboard scores were collected for a representative subset of 100 models per competition rather than for all 1,000 trained models per competition due to Kaggle submission limits and rate throttling. Second, the study focused only on completed Kaggle tabular binary classification competitions, so the findings may not generalise to multiclass classification, regression, time series, image, or natural language competitions. Third, only stratified 5-fold cross-validation was used as the main validation strategy. Repeated cross-validation or nested cross-validation could provide additional insight into validation stability but would require substantially more computation. Finally, the private leaderboard was treated as the closest available proxy for true

TABLE VI
BAYESIAN POSTERIOR SUMMARY BY COMPETITION

Competition	Models	Posterior Mean	Posterior Std	Mean Posterior Uncertainty	Mean Private AUC	Mean CV AUC	Mean Public AUC
Homesite	100	0.956274	0.009911	0.000890	0.956401	0.955840	0.956981
Porto	100	0.626994	0.013658	0.000890	0.627521	0.626901	0.625845
Santander	100	0.856646	0.033072	0.000890	0.855992	0.858434	0.858785

generalisation performance, but it is still a finite hidden test split and may contain sampling noise.

B. Future Work

Future work could extend this research across more Kaggle competitions and domains. Additional validation strategies, such as repeated cross-validation, nested cross-validation, adversarial validation, and time-based validation splits, could also be compared. More advanced Bayesian models could estimate competition-specific and model-family-specific bias terms rather than using only global bias estimates. Future work could also analyse how submission behaviour affects validation reliability, including the number of submissions, submission order, and adaptive modelling decisions based on public leaderboard feedback.

VII. CONCLUSION

This study evaluated the reliability of cross-validation in Kaggle tabular binary classification competitions using a large-scale empirical experiment and Bayesian melding framework. Across three completed Kaggle competitions, 3,000 model configurations were trained and evaluated using stratified 5-fold cross-validation. A representative subset of 300 models was then scored on Kaggle public and private leaderboards.

The results show that cross-validation AUC was a highly reliable predictor of private leaderboard AUC in the studied competitions. The within-competition centred Pearson correlation between cross-validation AUC and private leaderboard AUC was 0.9958, while the centred Pearson correlation between public and private leaderboard AUC was 0.9988. The mean global cross-validation bias was also very small at 0.00042 AUC.

Competition-level differences were still observed. Homesite showed the smallest validation gap, Porto showed slightly pessimistic validation behaviour, and Santander showed mild optimism in both cross-validation and public leaderboard scores. This indicates that cross-validation reliability is strong overall but still depends on dataset characteristics and competition design.

Bayesian melding provided a useful framework for integrating cross-validation, public leaderboard, and private leaderboard scores as noisy observations of latent true model performance. Overall, the study suggests that carefully implemented stratified cross-validation can be highly reliable in Kaggle tabular binary classification competitions, while still requiring uncertainty-aware interpretation.

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